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B.TECH/ M.Tech (Integrated) DEGREE EXAMINATION, JUNE 2025

Third Semester

21CSS201T - COMPUTER ORGANIZATION AND ARCHITECTURE

(For the candidates admitted from the academic year 2021-2022 to 2024-2025)

Note:

(i) **Part - A** should be answered in OMR sheet within first 40 minutes and OMR sheet should be handed over to hall invigilator at the end of 40th minute.

(ii) **Part - B & Part - C** should be answered in answer booklet.

Time: 3 hours

Max. Marks: 75

PART - A (20 x 1 = 20 Marks)

Answer **ALL** Questions

Marks BL CO PO

- | | | | | |
|---|---|---|---|---|
| 1. How many bits are used to represent each decimal digit in BCD? | 1 | 1 | 1 | 1 |
| A) 2 bits | | | | |
| B) 3 bits | | | | |
| C) 4 bits | | | | |
| D) 8 bits | | | | |
| 2. Convert the binary number 0110011 into its 2's complement. | 1 | 2 | 1 | 2 |
| A) 1001000 | | | | |
| B) 1001101 | | | | |
| C) 1001100 | | | | |
| D) 0110011 | | | | |
| 3. High level design is exhibited in | 1 | 1 | 1 | 1 |
| A) Computer Architecture | | | | |
| B) Computer Organization | | | | |
| C) Computer BIOS | | | | |
| D) Computer Architecture and Organization | | | | |
| 4. Generate the gray code for the binary number 1010. | 1 | 4 | 1 | 2 |
| A) 1111 | | | | |
| B) 1001 | | | | |
| C) 1100 | | | | |
| D) 1000 | | | | |
| 5. The program should be in _____ memory at the time of execution. | 1 | 2 | 3 | 1 |
| A) Primary | | | | |
| B) Secondary | | | | |
| C) Floppy Disk | | | | |
| D) Pen drive | | | | |
| 6. The instruction Store R4, LOC will do which of the following? | 1 | 3 | 3 | 1 |
| A) copies the operand in memory location LOC to processor register R4 | | | | |
| B) copies the operand in processor register R4 to memory location LOC | | | | |
| C) copies the operand in memory location LOC | | | | |
| D) copies the operand in processor register R4 | | | | |
| 7. The control line of the Bus is | 1 | 3 | 3 | 1 |
| A) uni-directional | | | | |
| B) bi-directional | | | | |
| C) multi-directional | | | | |
| D) un-directional | | | | |
| 8. The instruction to be currently executed is stored in | 1 | 1 | 3 | 1 |
| A) MDR | | | | |
| B) IR | | | | |
| C) PC | | | | |
| D) MAR | | | | |

9. In booth recoding, the bits pair (1, 0) are recoded by
 A) 0 B) +1
 C) -1 D) -2
10. The mantissa is also called as
 A) Mnemonics B) Significand
 C) Symbols D) Numbers
11. The full adder can add a maximum of _____ bits.
 A) 1 B) 2
 C) 3 D) 0
12. The double precision of the IEEE floating point number can be represented by
 A) 16 bits B) 24 bits
 C) 32 bits D) 64 bits
13. The micro routine for all instructions are stored in special memory called
 A) Micro instructions B) Control store
 C) Instruction store D) Control word
14. The _____ signal permits the processor to hold off until the memory operation is completed.
 A) MFC B) WPMC
 C) WMFC D) ALB
15. The situation wherein the data of operands are not available is called ____
 A) Structural hazard B) Deadlock
 C) Stall D) Data hazard
16. The pipelining process is also called as
 A) Superscalar operation B) Assembly line operation
 C) Von Neumann Cycle D) Instruction Sequencing
17. MIMD stands for
 A) Multiple Instruction Multiple Decoding B) Multiple Instruction Multiple Data
 C) Multiple Information Multiple Decoding D) Multiple Information Multiple Data
18. The Von Neumann architecture is based on
 A) SIMD B) SISD
 C) MIMD D) MISD
19. The ARM core uses _____ architecture.
 A) EPIC B) CPIC
 C) RISC D) CISC
20. ARM processor core is of _____ bits.
 A) 24 B) 32
 C) 48 D) 128

PART - B (5 x 8 = 40 Marks)
Answer **ALL** the Questions

Marks BL CO PO

- 21 a. Convert the Decimal number 105.15 into Binary, Octal, Hexadecimal and BCD forms. 8 2 1 2
(OR)
b. Describe all the logic gates with symbols and truth table. 8 2 2 1
- 22 a. How byte and word addressing is done in Big Endian and Little Endian if the word length of the computer is 32 bits. Explain with the help of diagrams. 8 2 5 1
(OR)
b. Evaluate the arithmetic statement $X = (A - B) * (C - D)$ in zero, one, two and three address instruction formats. 8 2 5 2
- 23 a. Solve 60 x 40 using Carry-Save Addition of Summands. 8 3 3 1
(OR)
b. Express how the floating point numbers are represented by IEEE Standard. 8 3 3 1
- 24 a. Draw the Multiple Bus Organization and construct the control signal sequence for ADD R4, R5, R6. 8 3 4 1
(OR)
b. What is a data hazard? Discuss the side effects of data hazard and how to overcome them. 8 3 3 1
- 25 a. Explain Flynn's Classification and its four architectures in detail. 8 2 4 1
(OR)
b. Illustrate the Thumb Instruction Set and features of the ARM Processor. 8 2 4 1

PART - C (1 x 15 = 15 Marks)
Answer **ANY ONE** Questions

Marks BL CO PO

- 26 Classify the various Addressing modes and explain each type with an example. 15 4 5 1
- 27 Perform the division for the following number using both the Restoring Division method and the Non-Restoring Division Method: Divide 13 by 3, use binary arithmetic and show each step clearly. 15 4 3 1
